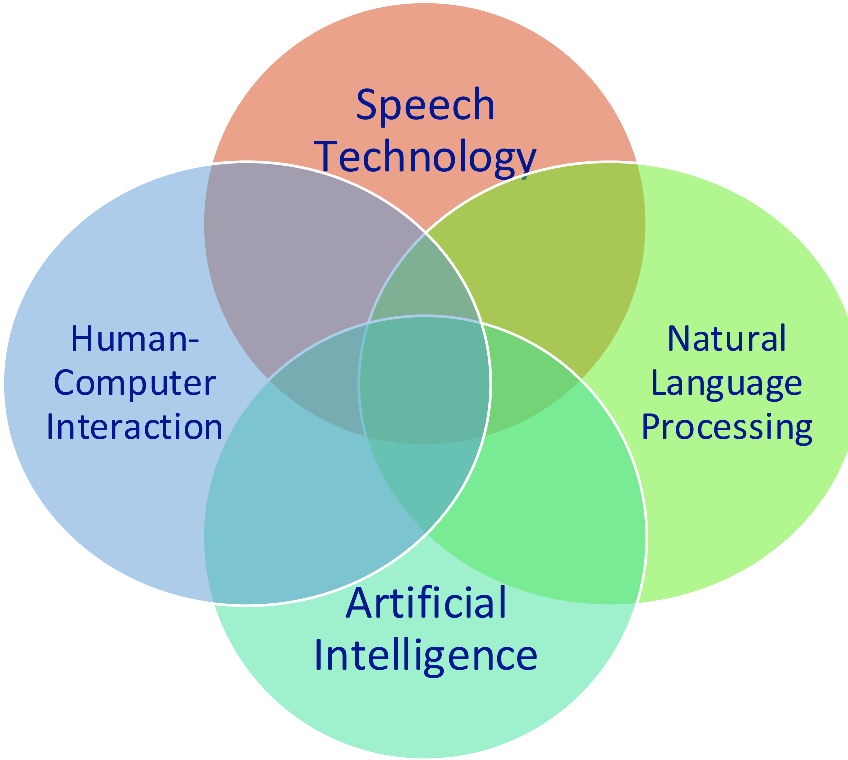
1. Introduction to Speech Technology

Nguyen Thi Thu Trang

Speech Technology

- Recognition, understanding and generation of human speech
- Why it Matters:
  - Communication efficiency in human-computer interaction
  - Accessibility for people with disabilities.



Applications of Speech Technology

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Applications of Speech Technology

- Virtual Assistant (e.g. Siri, Alexa)
- Speech Translation
- User Authentication
- Automated customer service systems
- Accessibility for people with disabilities, e.g. Screen Readers

Siri

Applications of Speech Technology

- **Healthcare**: Diagnosing Alzheimer's through speech patterns.
- **Education**: Language learning tools powered by TTS.
- **Accessibility**: Smart homes controlled via voice.
- **Entertainment**: Synthetic voices for personalized audiobooks.

Tasks of Speech Technology, e.g. TTS

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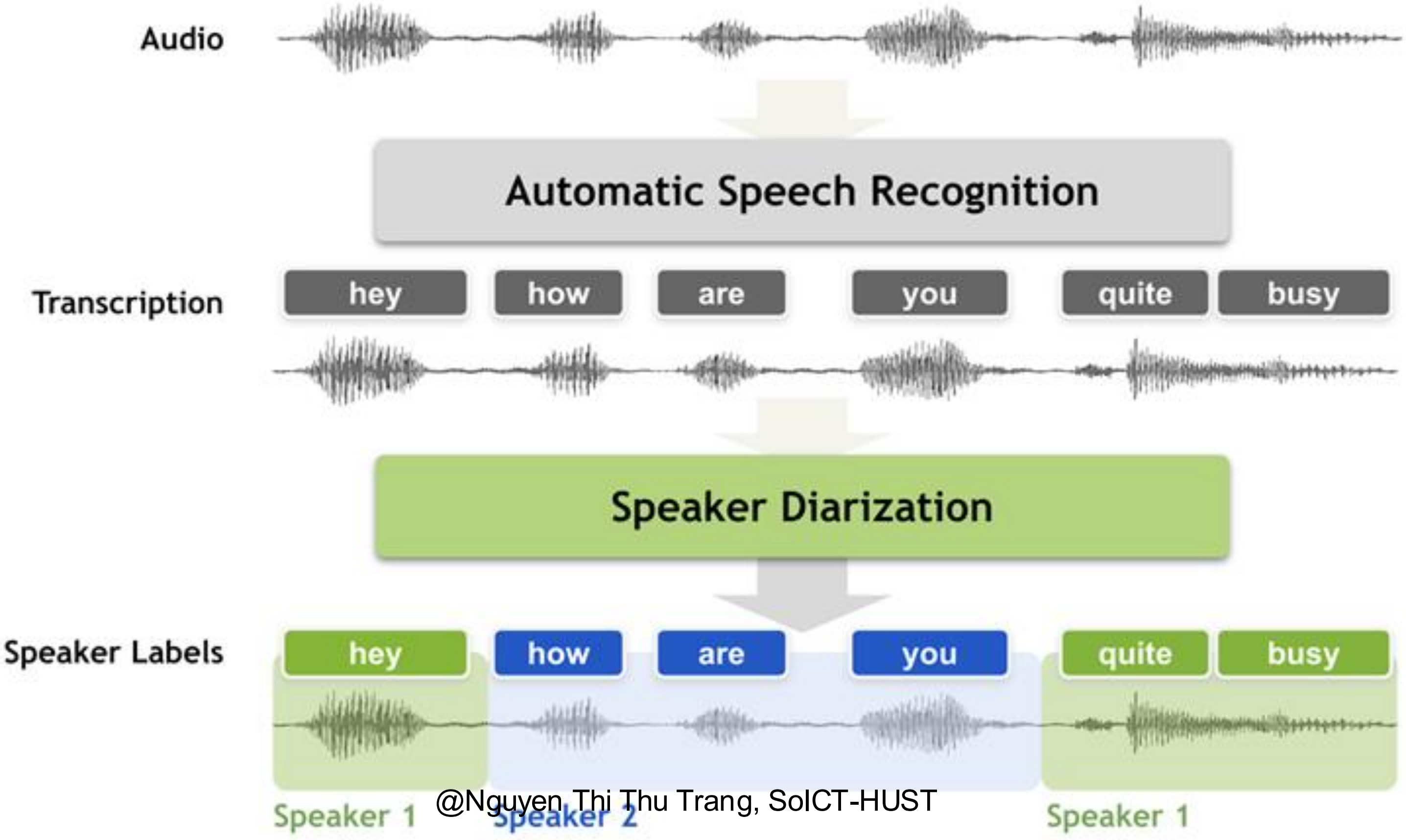
Speech Technology Tasks

- Speaker Recognition
- Speaker Diariazation
- Speech Emotion Recognition (SER)
- Automatic Speech Recognition (ASR)
- Spoken Language Understanding (SLU)
- Speech Synthesis / Text-To-Speech (TTS)
- Mispronunciation Detection and Diagnosis (MD&D)

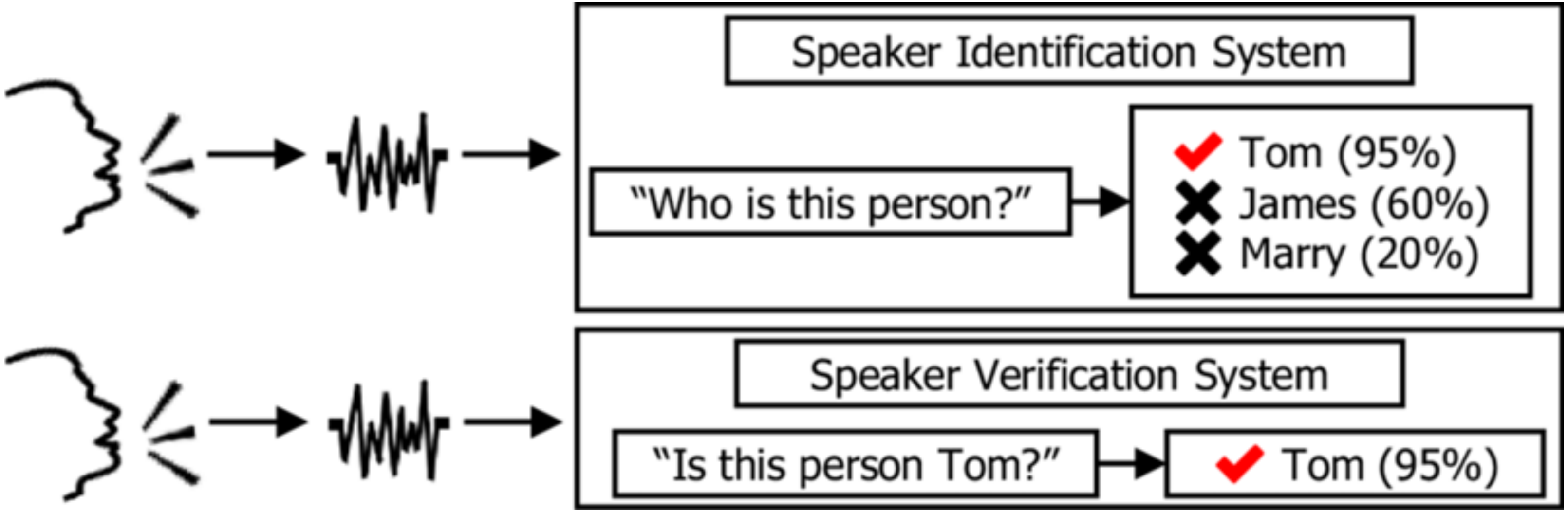
Speech Technology Tasks



E.g. Speech Transcription

Speech Technology Tasks

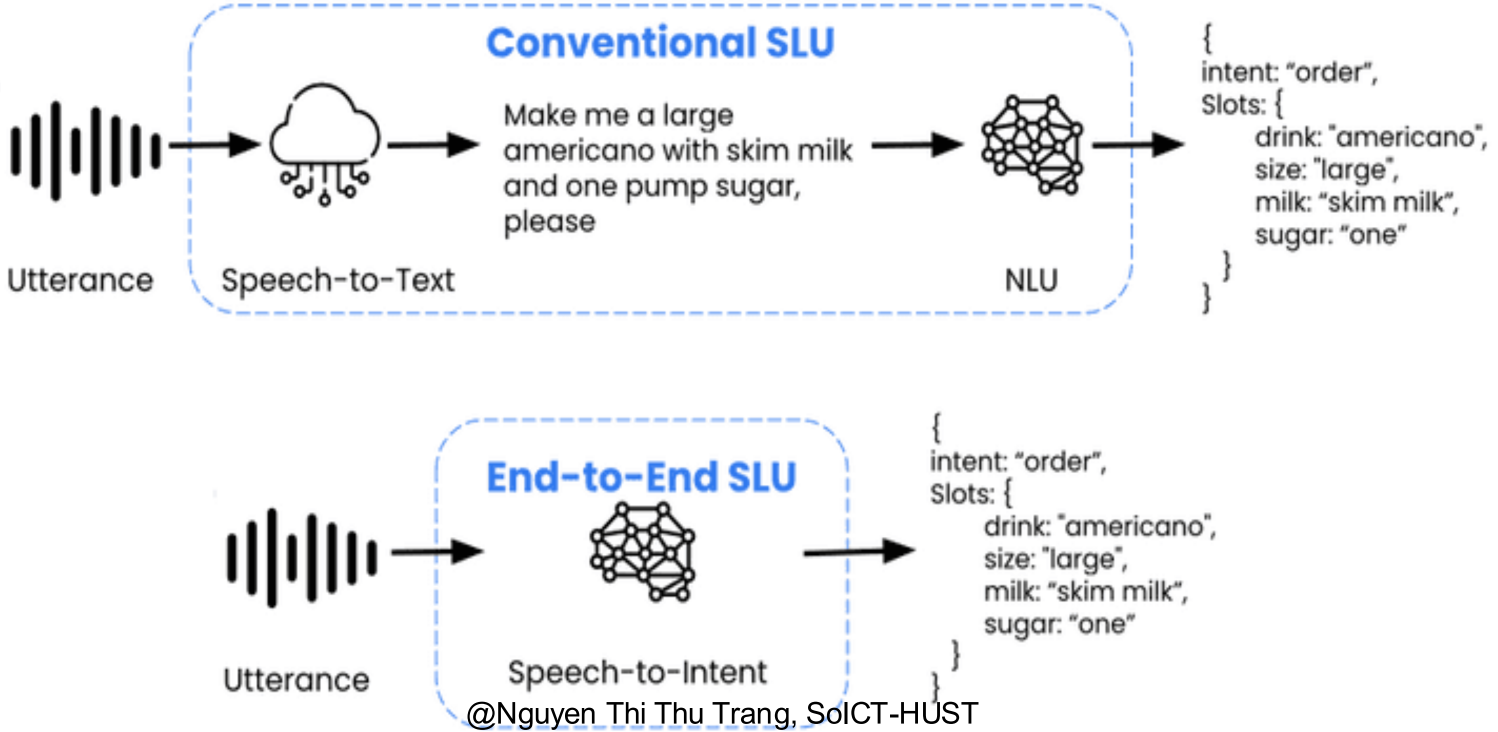
Speaker Recognition



Nuance Voice Biometrics: Customer Verification/Identification (2018)

Spoken Language Processing Tasks

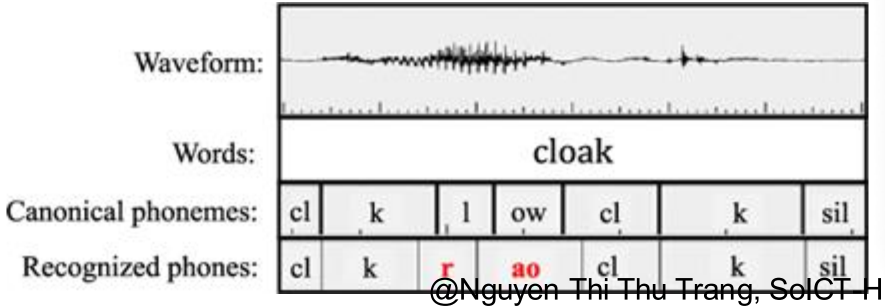
Spoken Language Understanding



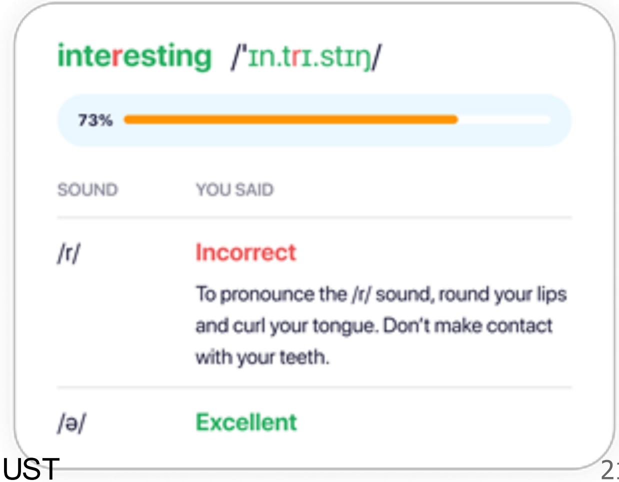
Google Assistant: Calls Local Businesses To Make Appointments (2018)

Spoken Language Processing Tasks

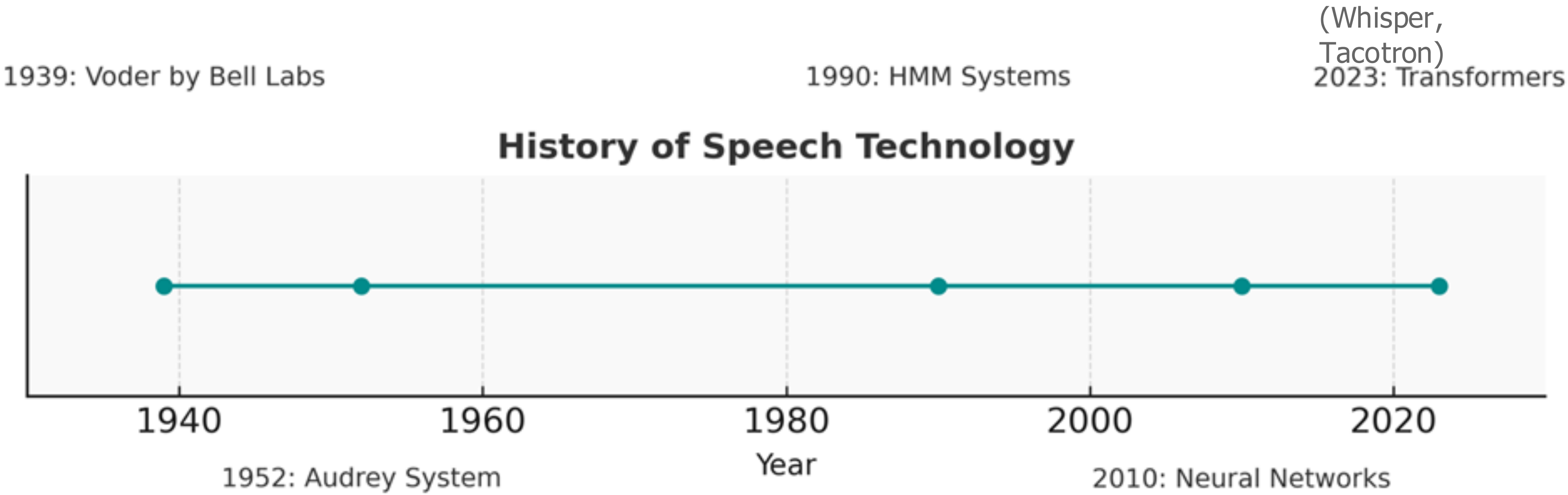
Mispronunciation Detection and Diagnosis (MD&D)



ELSA Speak



History of Speech Technology



(First automatic digit recognition)

Comparison of accuracy from early systems to modern ASR?

What is the breakthrough difference in accuracy between early ASR systems and modern models?

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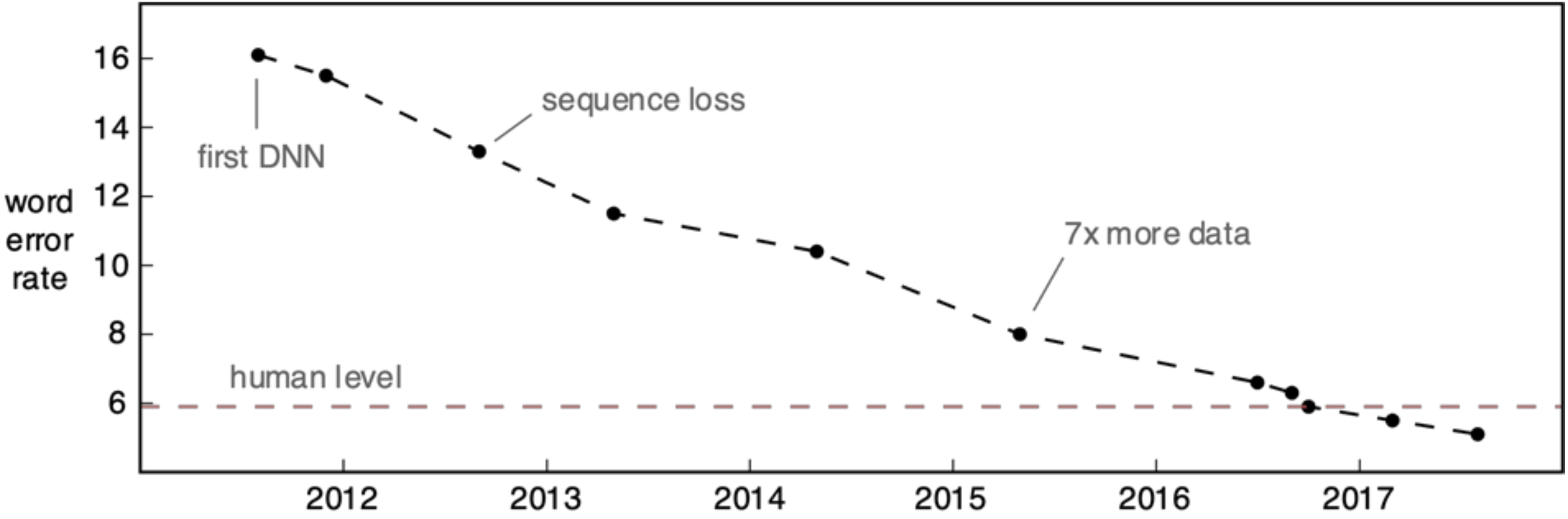
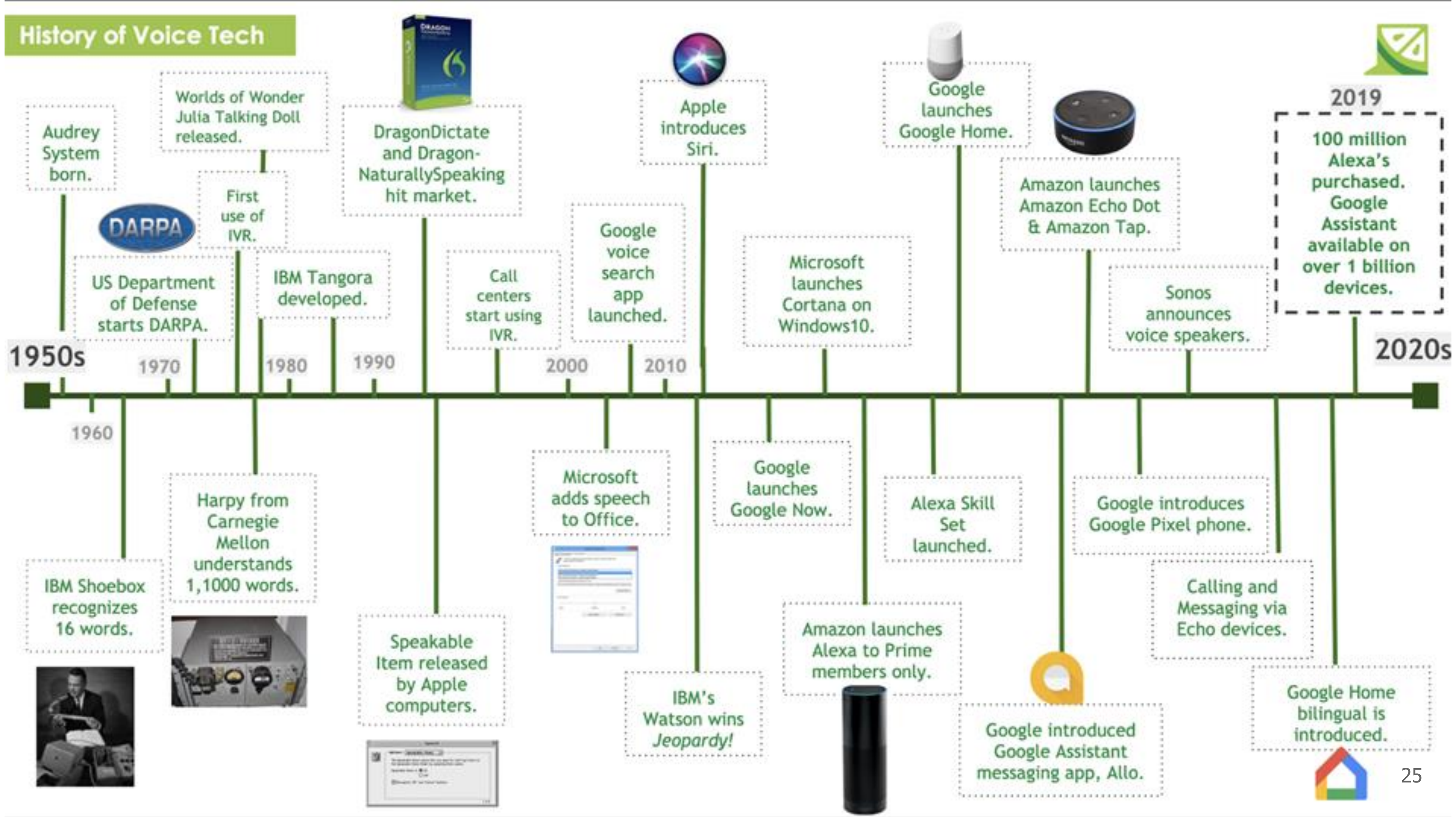
Challenges in ASR

Switchboard Conversational Speech Recognition (WER)



https://awni.github.io/speech-recognition/


History of Voice Tech
1950s
Audrey System born.
1960
IBM Shoebox recognizes 16 words.
1970
DARPA
US Department of Defense starts DARPA.
Worlds of Wonder Julia Talking Doll released.
First use of IVR.
Harpy from Carnegie Mellon understands 1,1000 words.
1980
IBM Tangora developed.
1990
DRAGON
DragonDictate and Dragon-NaturallySpeaking hit market.
Speakable Item released by Apple computers.
Call centers start using IVR.
2000
Microsoft adds speech to Office.
Google voice search app launched.
2010
Apple introduces Siri.
IBM's Watson wins *Jeopardy!*
Google launches Google Now.
Microsoft launches Cortana on Windows10.
Amazon launches Alexa to Prime members only.
Alexa Skill Set launched.
Google launches Google Home.
Google introduced Google Assistant messaging app, Allo.
Amazon launches Amazon Echo Dot & Amazon Tap.
Google introduces Google Pixel phone.
Sonos announces voice speakers.
Calling and Messaging via Echo devices.
Google Home bilingual is introduced.
2019
100 million Alexa's purchased. Google Assistant available on over 1 billion devices.
2020s

Current Trends in Speech Technology

- **Deep Learning Dominance:**
  - End-to-end ASR systems outperforming traditional methods.
- **Real-Time Applications:**
  - On-device processing for privacy (e.g., offline voice commands).
- **Multimodal Systems:**
  - Combining speech with vision for enhanced understanding (e.g., lip-reading systems).
- **Ethical AI:**
  - Bias mitigation, responsible voice cloning policies.

Challenges in Speech Technology

- **Noise and Variability:**
  - Real-world environments vs. controlled datasets.
  - **Solution:** Data augmentation techniques.
- **Accents and Dialects:**
  - Underrepresentation in training data.
  - **Solution:** Collecting diverse datasets (e.g., Common Voice).
- **Data Scarcity:**
  - Low-resource languages facing neglect.
  - **Solution:** Transfer learning and synthetic data.
- **Ethical Concerns:**
  - Deepfake risks and privacy violations.
  - **Solution:** Regulation and ethical guidelines.

Pakistan’s former prime minister is using an AI voice clone to campaign from prison

/ Imran Khan’s party crafted a four-minute message using a tool from the AI firm ElevenLabs.

By Amrita Khalid, one of the authors of audio industry newsletter Hot Pod. Khalid has covered tech, surveillance policy, consumer gadgets, and online communities for more than a decade.

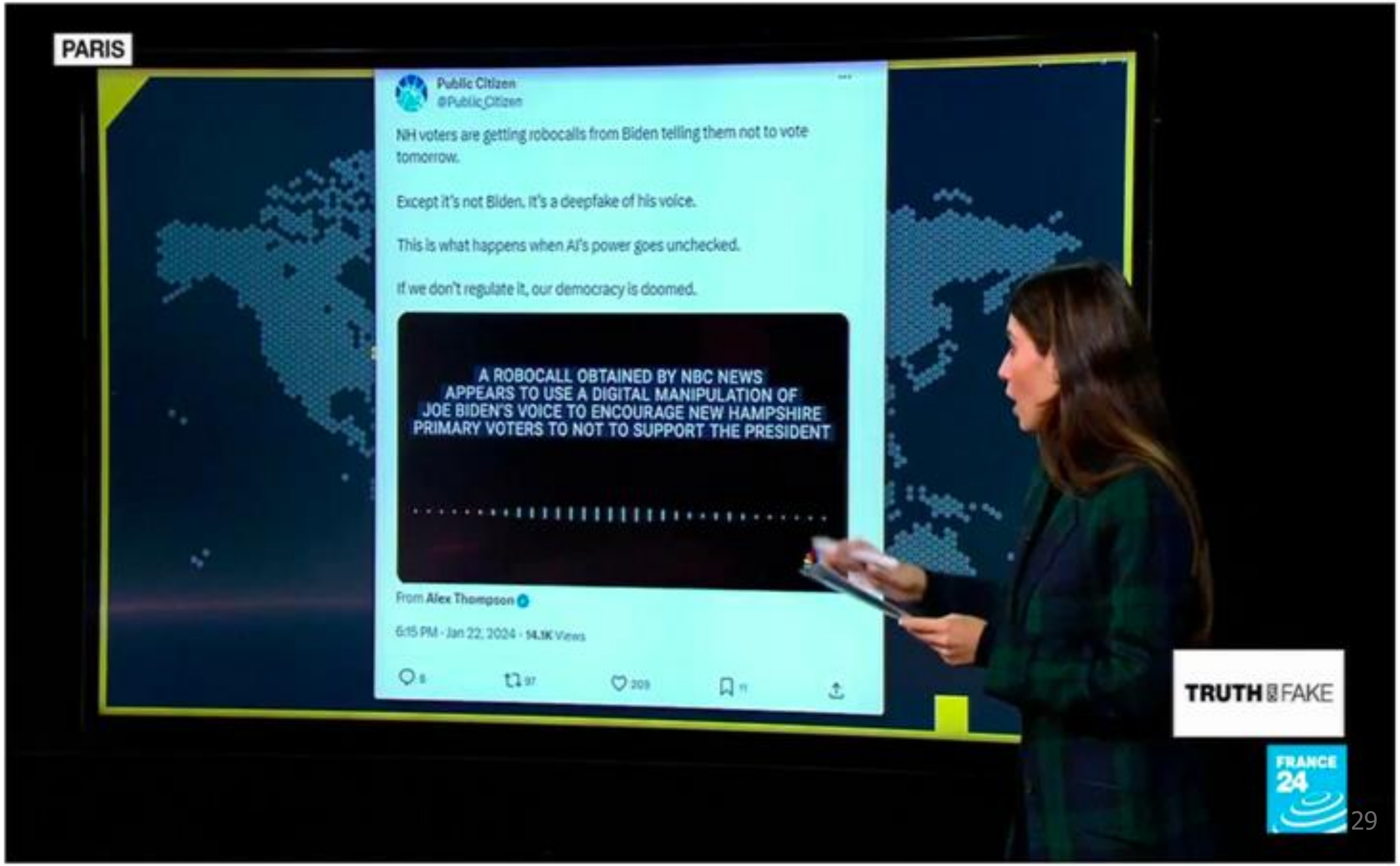
Dec 18, 2023, 2:41 PM PST

1 Comment (1 New)

The Biden Deepfake Robocall Is Only the Beginning

An uncanny audio deepfake impersonating President Biden has sparked further fears from lawmakers and experts about generative AI's role in spreading disinformation.

A New Era of Spoken Language Applications and Impact



Future Directions

- **Universal Models:**
  - Speech systems that generalize across languages and domains.
- **Emotion Recognition:**
  - Enhanced user interaction.
- **Low-Power Models:**
  - Expanding access to speech technology in remote areas.

Vision: Democratizing speech technology to make it accessible and ethical for everyone.

Conversational AI on device (Vbee 2024)

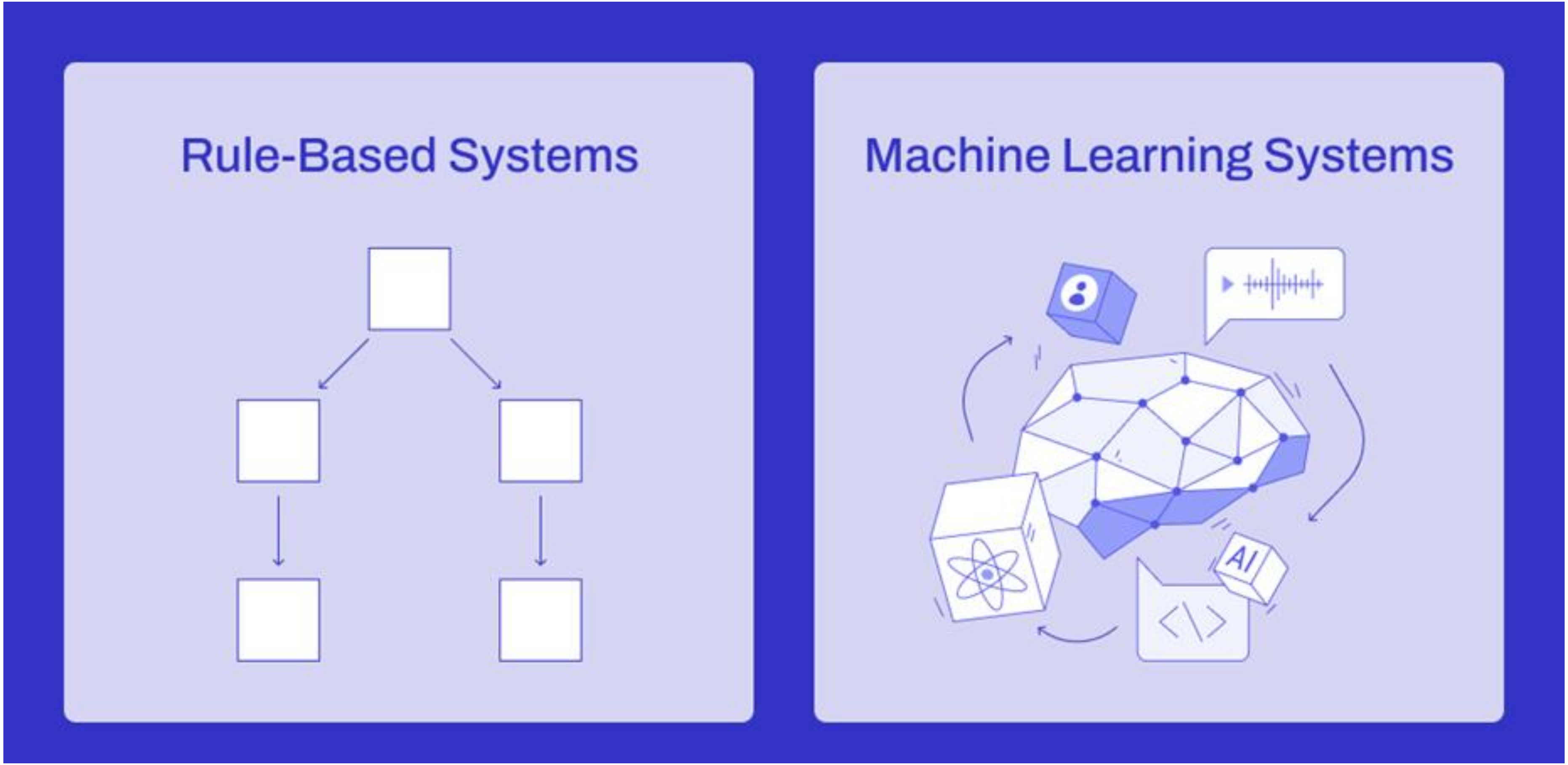
2. Linguistics and Phonetics

Nguyen Thi Thu Trang

What is the main difference between

Rule-based AI and

Machine Learning?



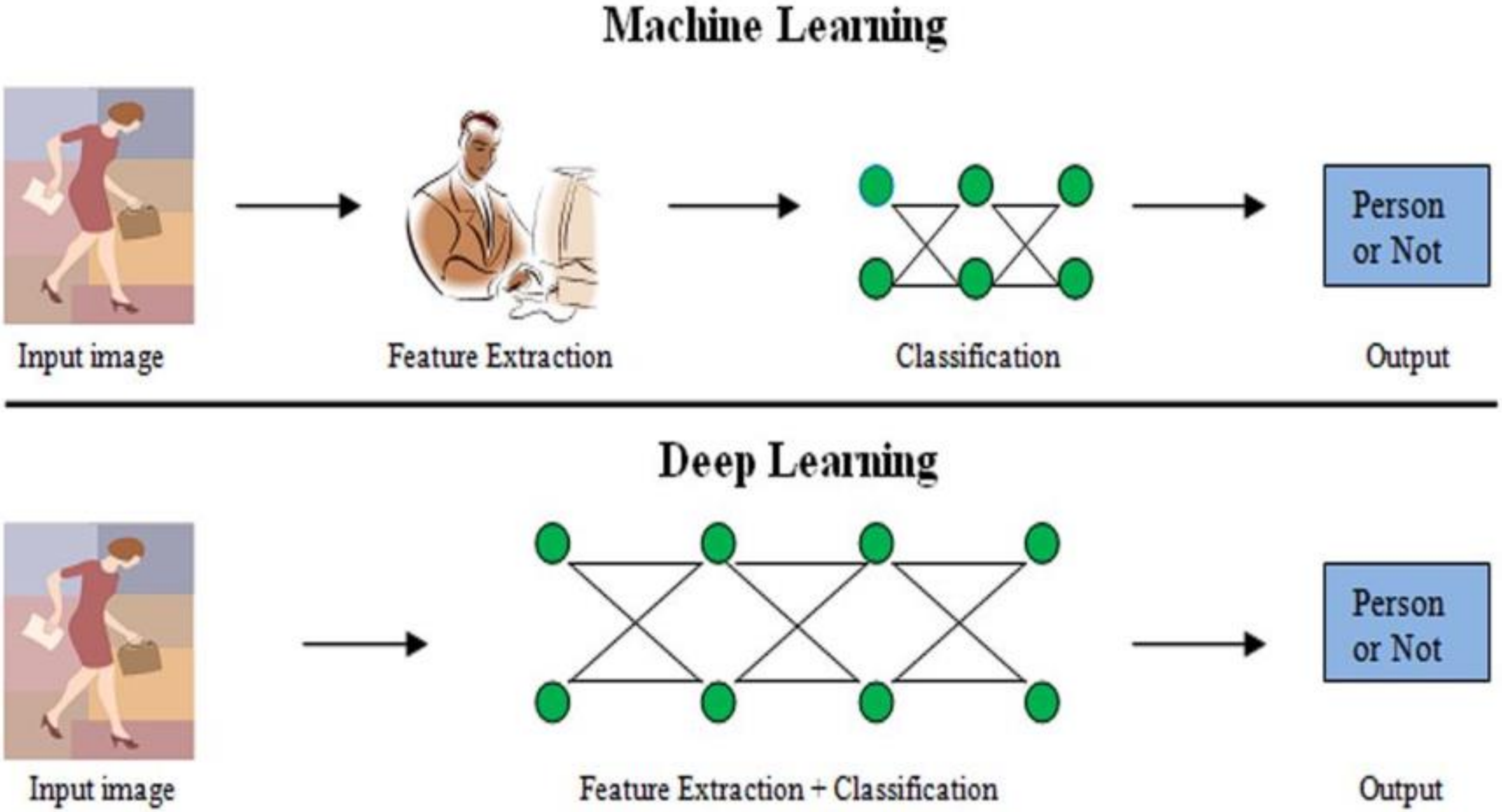
Predefined business rules and business logic by experts

Learns patterns from large datasets over time

What is the main difference between

Machine Learning and

Deep Learning?


Machine Learning
Person or Not
Input image
Feature Extraction
Classification
Output
Deep Learning
Person or Not
Input image
Feature Extraction + Classification
Output

Phonology vs. Phonetics

- **Sound systems** of a language.
- How **sounds function** in a particular language or languages.
- **Abstract, mental representations of sounds**.

- **Physical production** and **perception** of speech sounds.
- **Actual acoustic and articulatory properties** of sounds.
- **Universal properties of speech sounds**, not language-specific rules

Why Phonology and Phonetics?

Do we need phonology and phonetics to build systems that accurately process spoken language?

- Modern systems (based on deep learning) are far less reliant on encoding phonetic domain knowledge directly than previous approaches
- Allowing deep learning models to learn letter-sound mappings from data can perform much better than hand engineering phonetic structure into a recognition or synthesis system

Why Phonology and Phonetics?

However ...

- Basic understanding of phonology and phonetics and speech production helps with describing and debugging spoken language systems
  - E.g. how does an accent/tone change the sound of pronunciations?
- Phonetic categories are not arbitrary. They model the biology of how humans produce speech
  - Understanding the space of possible speech sounds gives a nice perspective on comparing spoken languages across the world, and how they evolve

What is Phonetics?

- **Definition:**
  - The study of speech sounds, their production, transmission, and perception.
- **Three Main Branches:**
  - **Articulatory Phonetics:** How speech sounds are produced.
  - **Acoustic Phonetics:** Physical properties of speech sounds.
  - **Auditory Phonetics:** How speech sounds are perceived by the listener

Example: Different between /b/ and /p/ sounds

Which parts of the body are involved in producing speech?

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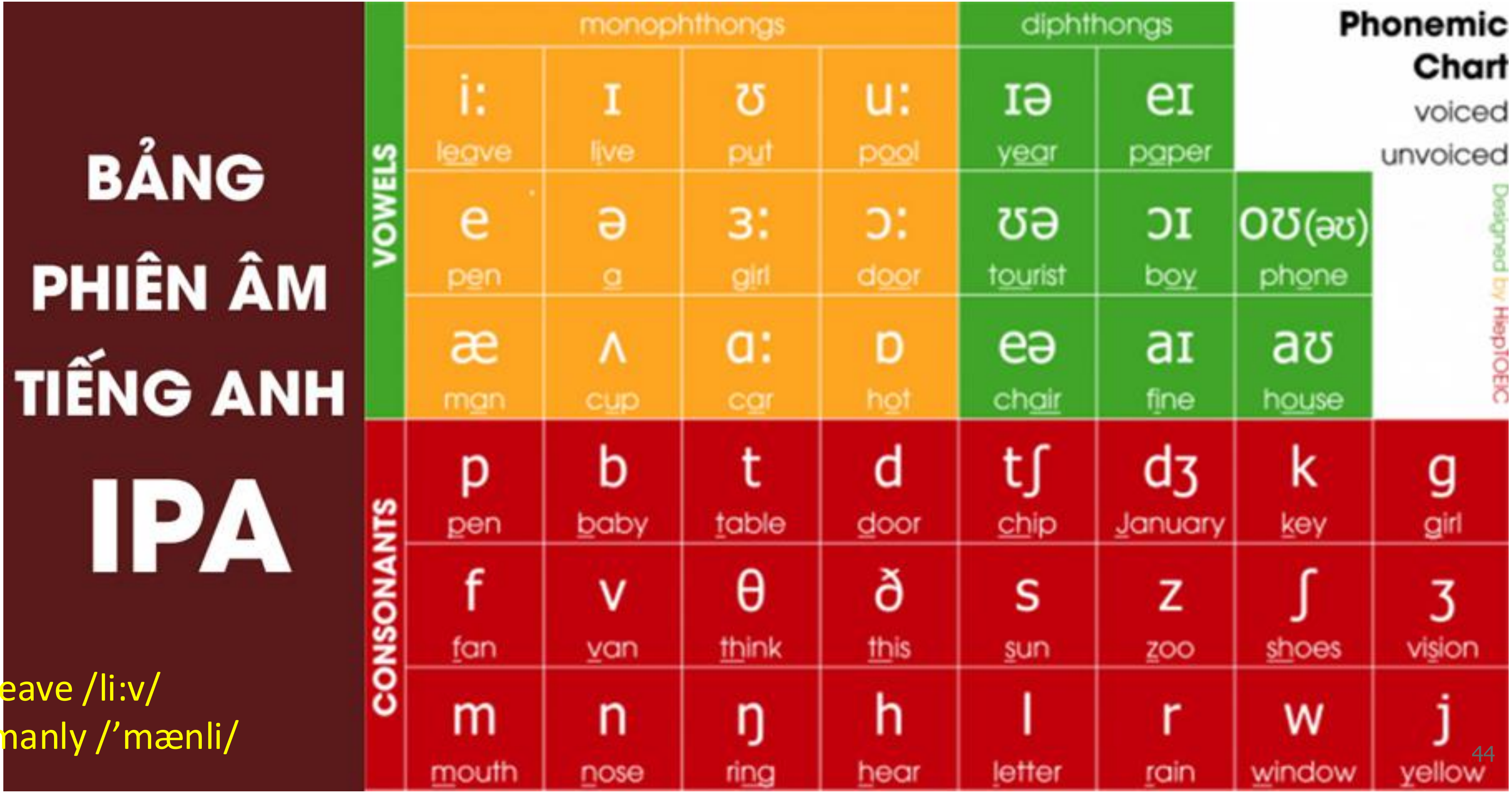
Letter vs. Sound

Grapheme vs. Phoneme

- **Grapheme**
  The smallest written unit representing a sound in speech.
  Examples: "d", "th", "ngh", "a", "au".
- **Phoneme**
  The smallest unit of sound in speech.
  Examples: /d/, /th/, /ŋ/, /a/, /ăw/.

IPA (International Phonetic Alphabet)



leave /li:v/

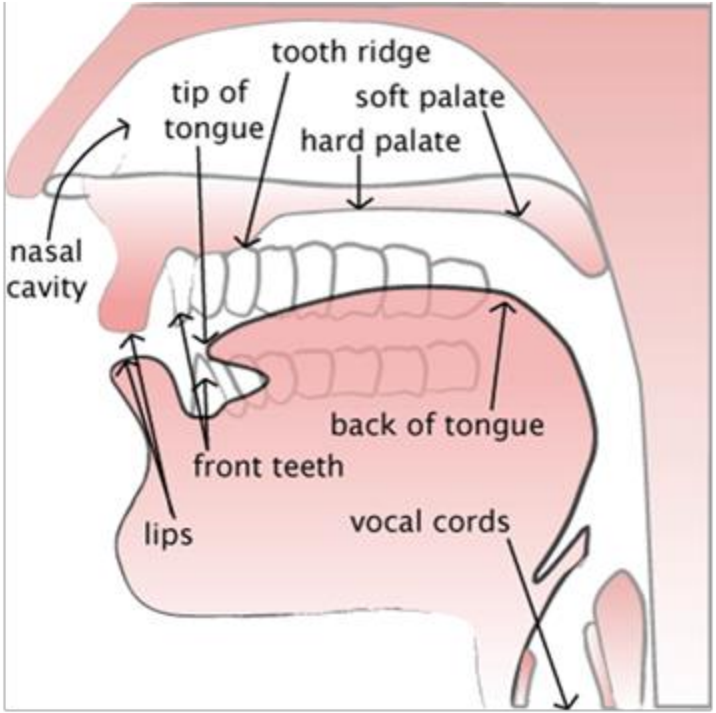
manly /’mænli/

Speech Production

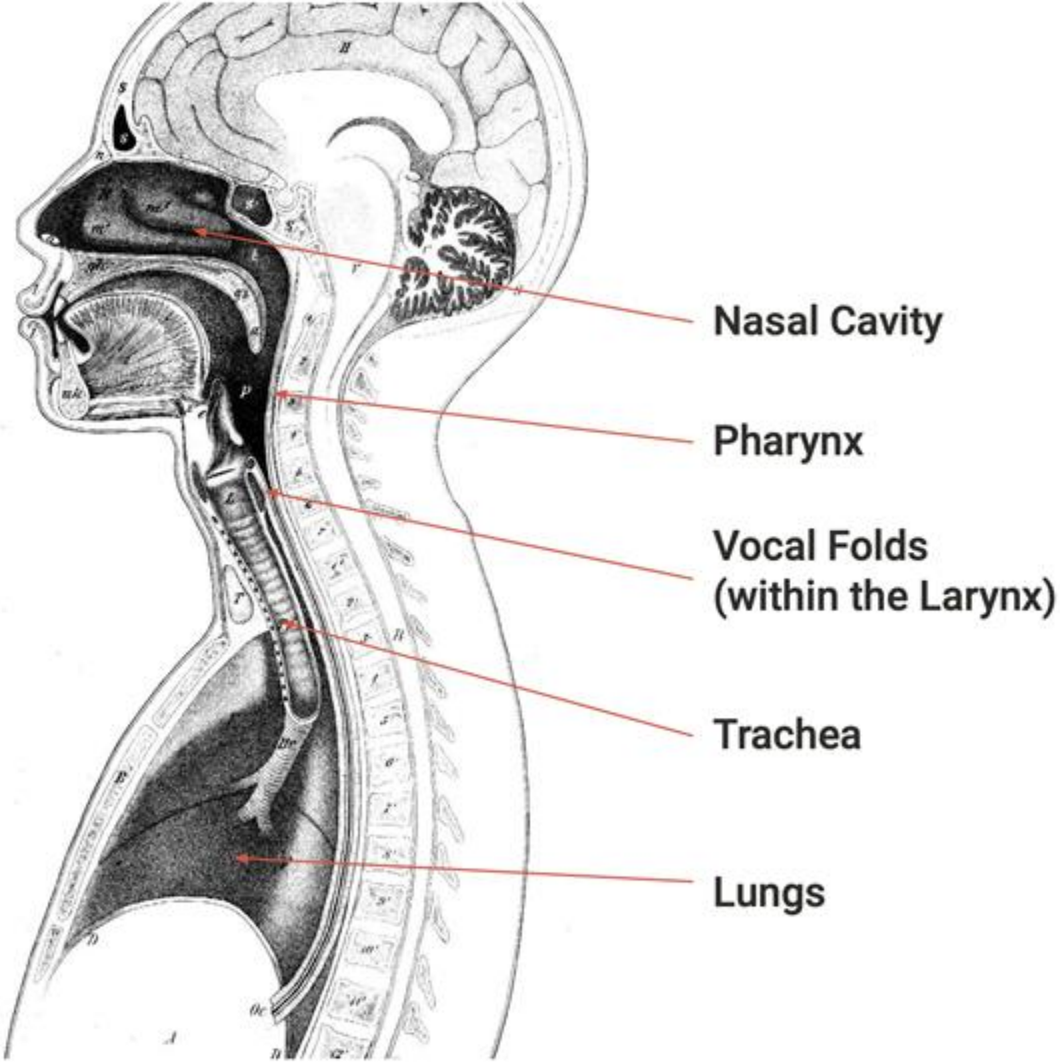
- **Flow**: we (normally) speak while breathing out. “Pulmonic egressive airstream” (Luồng khi từ phổi)
  - Airstream sets vocal folds in motion. Vibration of vocal folds (rung dây thanh âm) produces sounds.
- **Resonance** (Điều chỉnh âm thanh): shape of vocal tract causing harmonics
- **Articulation** (Phát âm): manipulation of airflow
  - Oral tract (Đường miệng): uvula (lưỡi gà), soft palate (velum - vòm miệng mềm), hard palate, tongue, lips, teeth
  - Nasal tract (Đường mũi)

Làm thế nào để tạo ra tiếng nói?

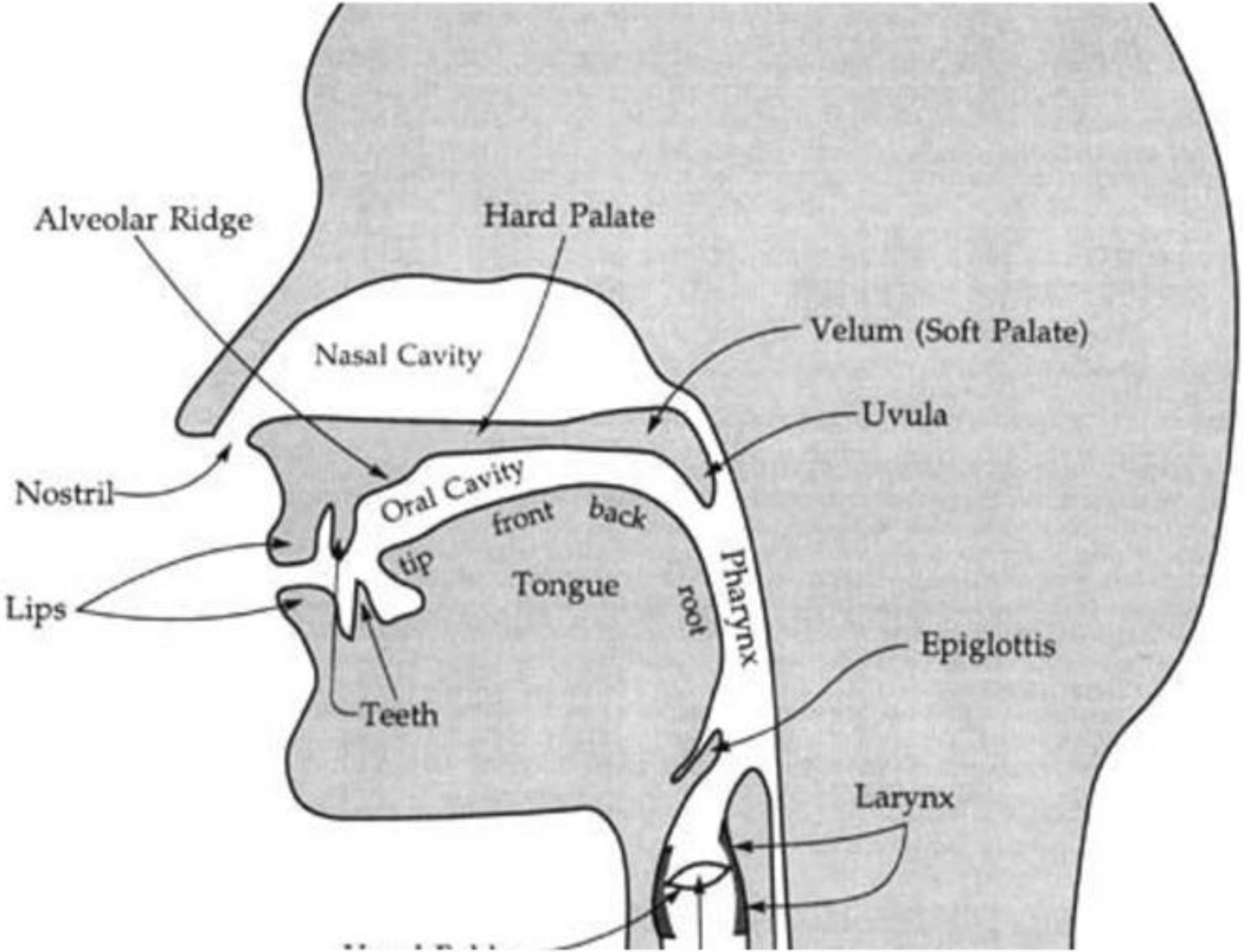
- **Hít vào và chuẩn bị không khí:** Phổi hít vào và điều chỉnh luồng khí qua cơ hoành (diaphragm).
- **Đẩy không khí qua thanh quản:** Dây thanh âm rung tạo âm thô.
- **Hình thành âm thanh:** Lưỡi, môi và khoang miệng định hình âm thanh.
- **Điều chỉnh âm rõ ràng:** Phối hợp lưỡi, môi, răng và hàm để



Sagittal section of the vocal tract

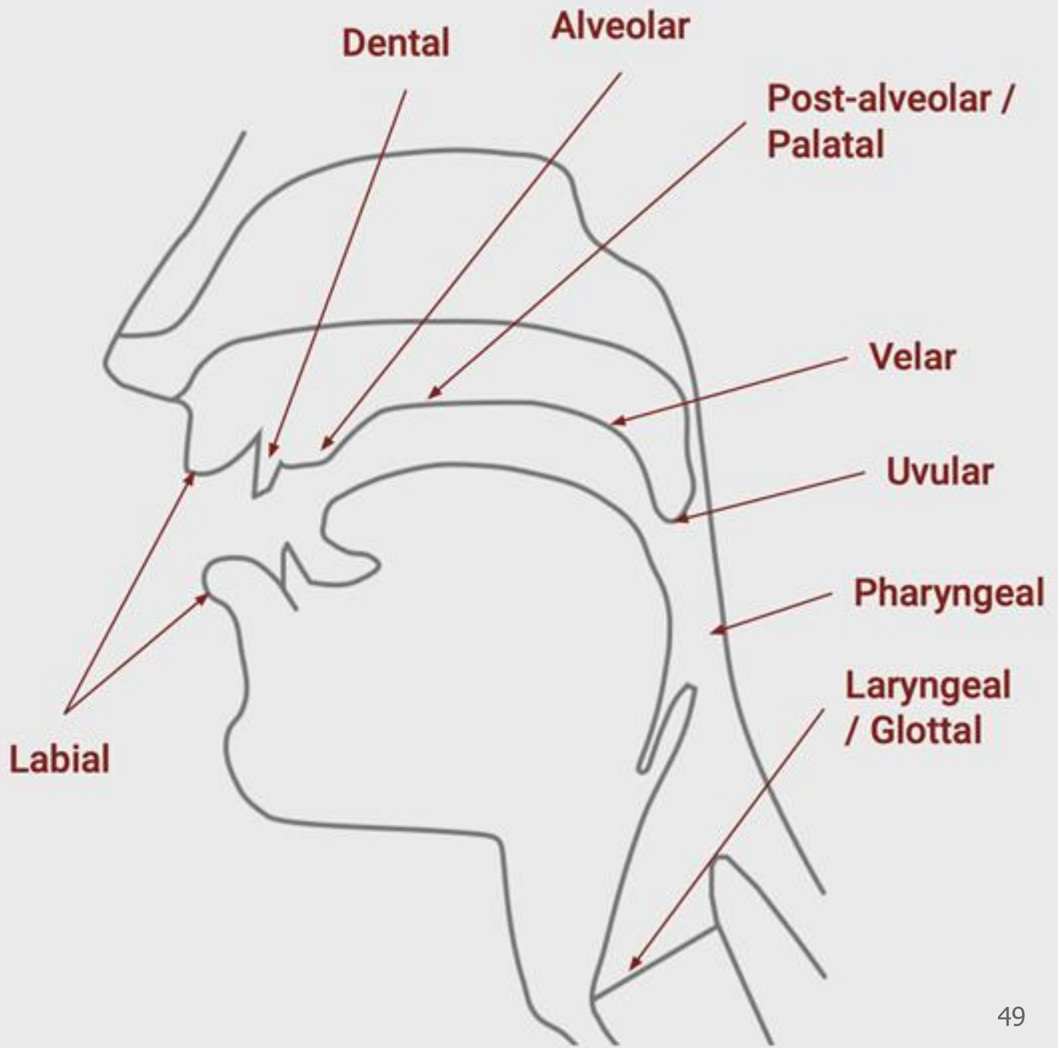


Sagittal section of the vocal tract



Place of Articulation

- **Consonants** are classified according to the location where the airflow is most constricted
  - This is called **place of articulation**
- Three major kinds of place articulation:
  - **Labial** (with lips)
  - **Coronal** (using tip or blade of tongue)
  - **Dorsal** (using back of tongue)



THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

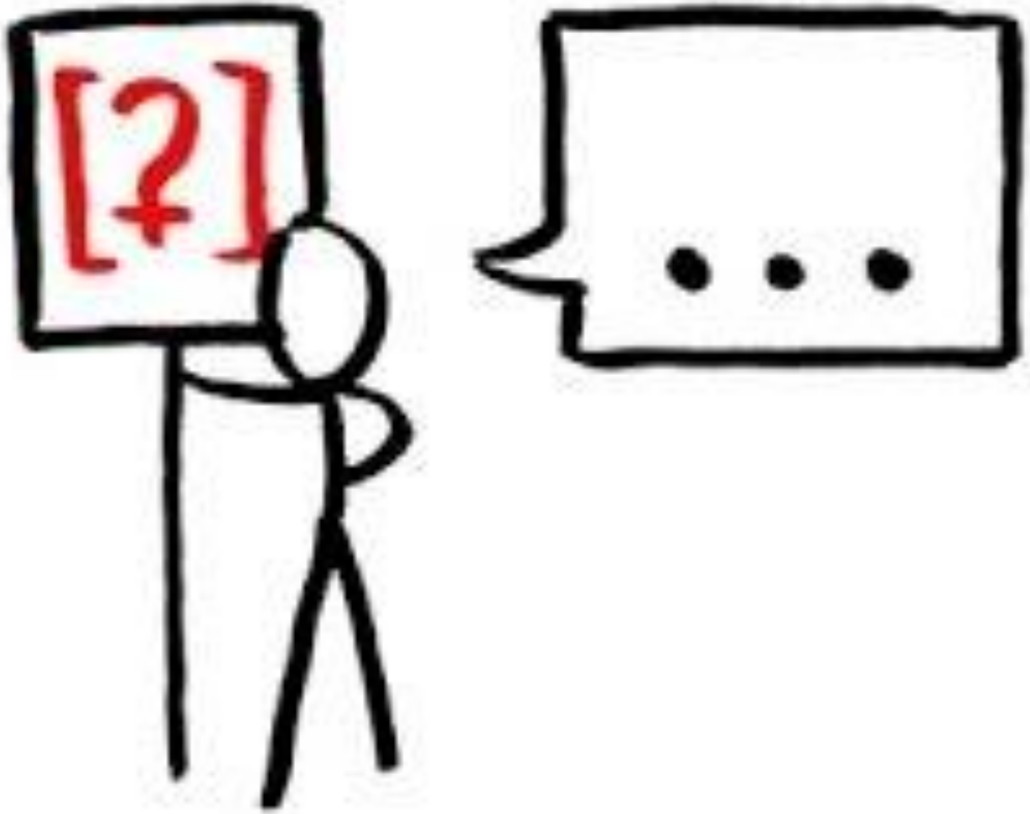
CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

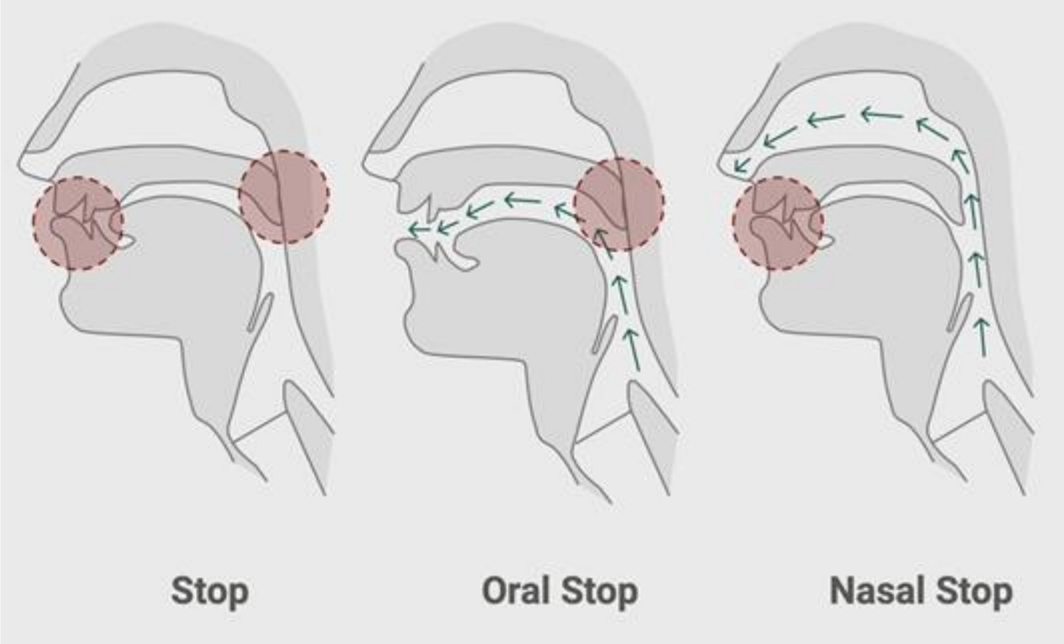
Place of Articulation

Place of Articulation

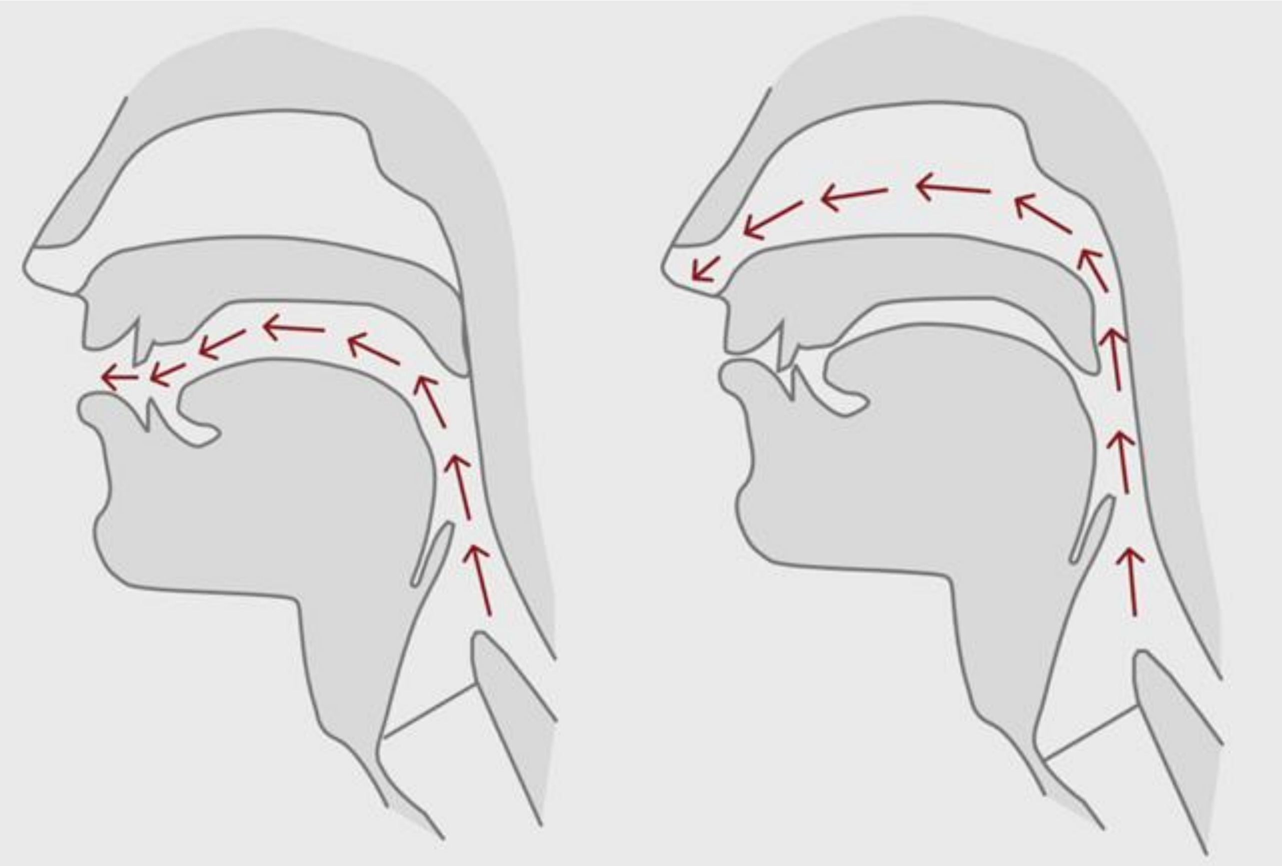


Manner of Articulation

- **Stop (Âm tắc):** complete closure of articulators, so **no air** escapes **through mouth**
- **Oral stop (Âm tắc miệng):** palate is raised, **no air** escapes **through nose**. Air pressure builds up behind closure, explodes when released
  - p, t, k, b, d, g
- **Nasal stop (Âm tắc mũi):** oral closure, but palate is lowered, **air escapes through nose**
  - m, n, ng

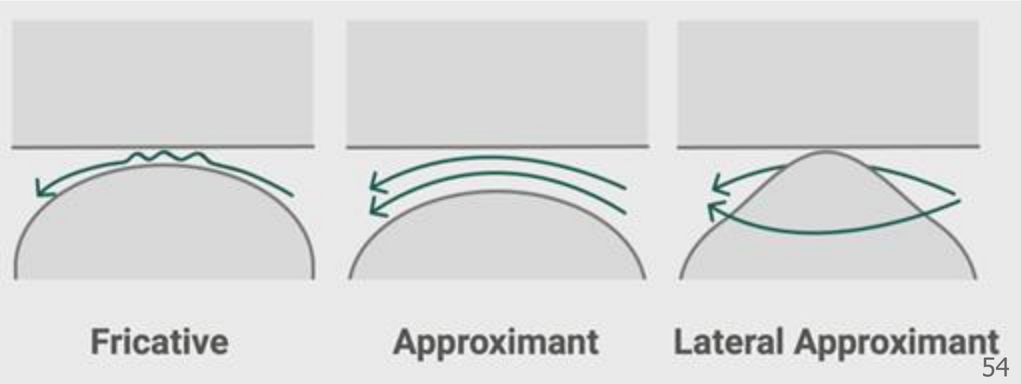


Oral vs. Nasal Sounds



Articulation Manner of Consonants

- **Fricatives (Âm xát)**: close approximation of two articulators, resulting in turbulent airflow between them, producing a hissing sound
  - f, v, s, z, th, dh
- **Approximant (Âm xấp)**: not quite-so-close approximation of two articulators, so no turbulence
  - y, r
- **Lateral approximant (Âm xấp bên)**: obstruction of airstream along center of oral tract, with opening around sides of tongue
  - l



Manner of Articulation

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	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

Manner of Articulation

Voiced vs. Unvoiced Consonants

Presence or absence of vocal cord vibration during the articulation of consonants.

- Vocal cords **vibrate** during pronunciation.
- Air from the lungs causes the vocal cords to rapidly open and close, producing sound waves.
- **Examples:**
  - **/b/** as in *bat*: vocal cords vibrate.
  - **/d/** as in *dog*.
  - **/z/** as in *zoo*.
  - **/v/** as in *van*.

- Vocal cords **do not vibrate** during pronunciation.
- Sound is produced solely by the airflow and contact between the articulators without vocal cord vibration.
- **Examples:**
  - **/p/** as in *pen*: vocal cords do not vibrate.
  - **/t/** as in *top*.
  - **/s/** as in *snake*.
  - **/f/** as in *fish*.

Voiced and Unvoiced Consonants

THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

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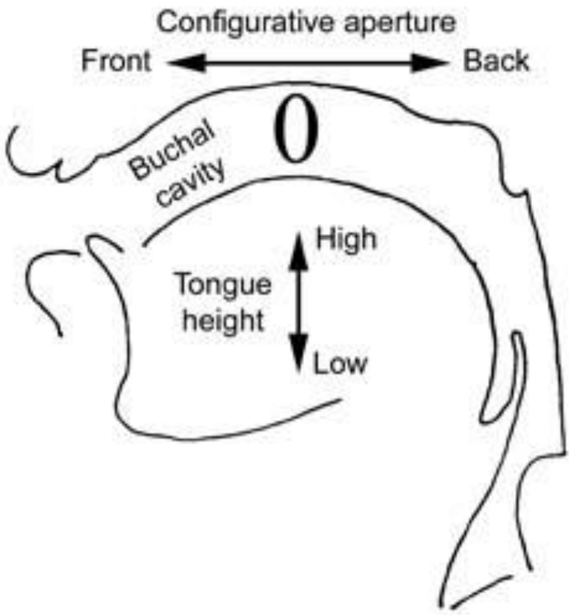
Unvoiced: Left, Voiced: Right

Tongue Position of Vowels: Tongue Height

How high or low the tongue is in the mouth when producing a vowel:

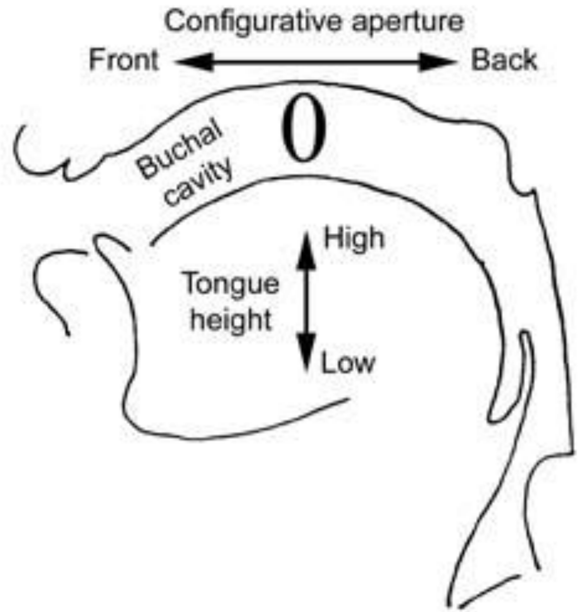
- **High vowels:** The tongue is raised close to the roof of the mouth.
  - Examples: ***/i/*** (*see*), ***/u/*** (*blue*).
- **Mid vowels:** The tongue is positioned midway between high and low.
  - Examples: ***/e/*** (*bed*), ***/o/*** (*go*).
- **Low vowels:** The tongue is lowered towards the bottom of the mouth.
  - Examples: ***/æ/*** (*cat*), ***/ɑ/*** (*father*).



Tongue Position of Vowels: Tongue Backness

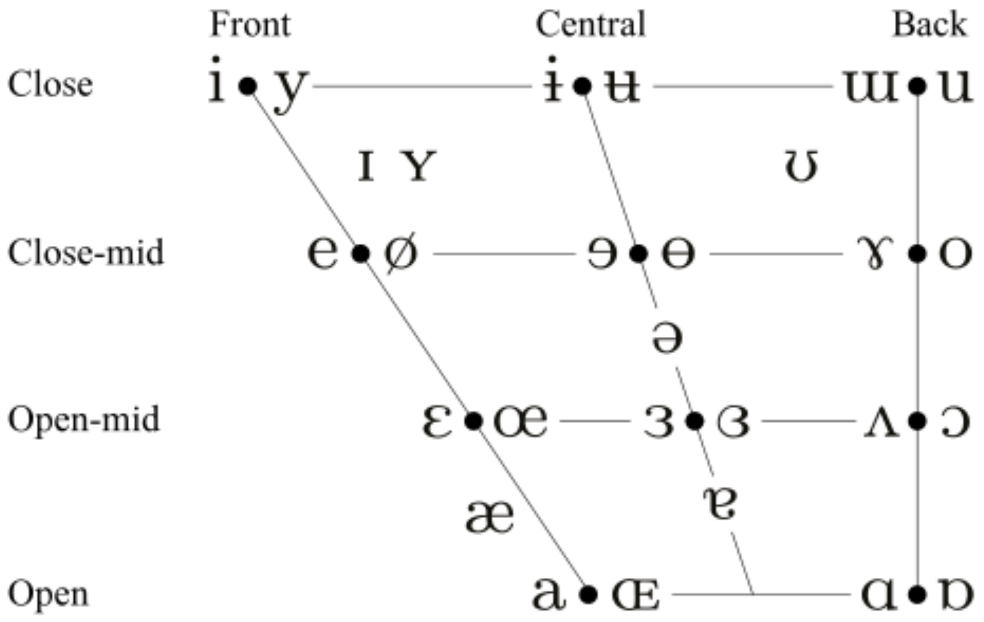
How far forward or backward the tongue is in the mouth:

- **Front vowels:** The tongue is positioned towards the front of the mouth.
  - Examples: ***/i/*** (*see*), ***/e/*** (*bed*).
- **Central vowels:** The tongue is centrally positioned.
  - Examples: ***/ə/*** (*about*), ***/ʌ/*** (*cup*).
- **Back vowels:** The tongue is positioned towards the back of the mouth.
  - Examples: ***/u/*** (*blue*), ***/o/*** (*go*).



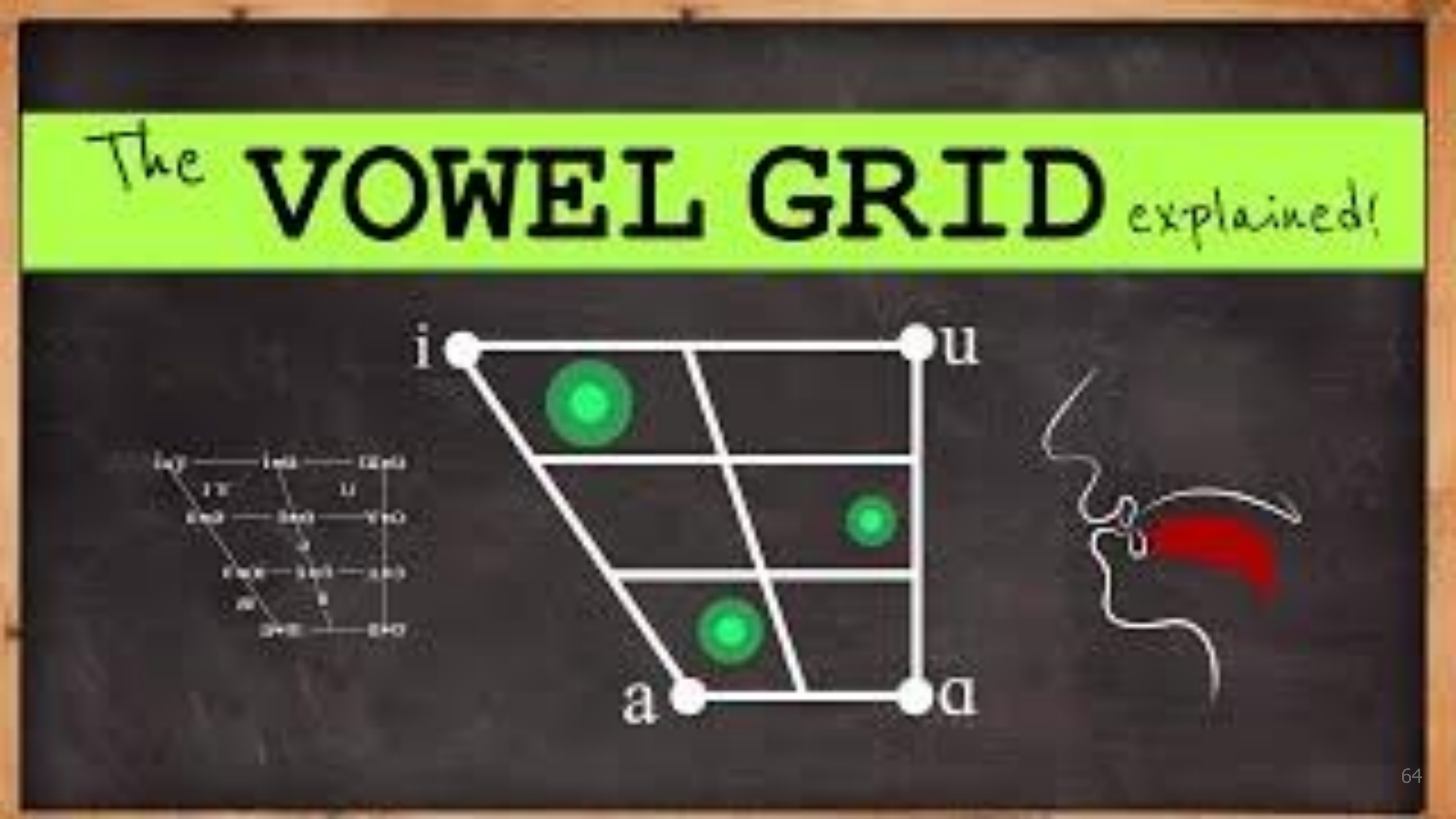
THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

VOWELS



Where symbols appear in pairs, the one
to the right represents a rounded vowel.

AEIOU


The VOWEL GRID explained!
i
u
a
ɑ
u